

EcoTrek: An AI-Driven Citizen-Centric Framework for Household Waste Classification, Public Waste Detection, and Collection Route Optimization

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Abstract

The management of municipal solid waste remains one of the most persistent demanding situations confronting unexpectedly urbanizing areas, with international waste technology projected to attain 2.2 billion heaps by 2025 and more or less 70% of the sector's population predicted to live in towns by 2050. The dominant reaction inside the literature, the Internet-of-Things (IoT)-enabled clever garbage bin (SGB), retrofits character boxes with ultrasonic, load, and fuel sensors, which involves recurring hardware and preservation fee and leaves informal or open-region dumping completely outside its subject of view. This paper presents EcoTrek, a software-first, synthetic-intelligence-pushed waste control framework that treats an everyday telephone digicam as the primary sensing modality in the area of committed per-bin hardware. EcoTrek bifurcates the citizen's reporting flow into a Household Waste Module, which classifies waste captured at domestic and recommends a context-appropriate reuse, recycling, or disposal action, and an Outdoor/Public Waste Module, which detects and geotags open dumping by way of type, quantity, and severity for municipal motion. Reports from each module converge on an administrative dashboard that applies spatio-temporal clustering to repeated, proximate reviews to become aware of series hotspots, which can be then sequenced by a concern- and capacity-conscious path-optimization engine. Building on an in-depth review of 4 primary sources, a foundational sensor-based clever-bin design[1], a 173-look at systematic literature review of IoT-enabled clever waste control[2], a LoRa-and-TensorFlow-based totally smart bin[3], and a lightweight Wi-Fi/cloud municipal system[4], together with the wider computer-vision and operations-research literature, this paper info EcoTrek's layered architecture, its phased development method, and the design decisions that reply immediately to the studies gaps identified inside the systematic review. Critically, the Household Waste Module's center classifier has been applied and empirically evaluated: a transfer-studying pipeline built on EfficientNetB0[9] became skilled on a

2,884-image, 4-class waste dataset[22] (Hazardous, Non-Recyclable, Organic, Recyclable) and evaluated on a held-out test set of 434 pictures, reaching an average test accuracy of 74.88% and a weighted F1-rating of zero.7483, with in keeping with-magnificence F1-rankings ranging from zero.6742 (Non-Recyclable) to zero.8088 (Hazardous). These outcomes, collectively with a proposed evaluation protocol for the outside detection, hotspot-detection, and course-optimization modules, are presented as evidence that a hardware-independent, AI-imaginative and prescient method can meaningfully supplement sensor-instrumented clever-bin systems in extending citizen-level waste control from family recycling decisions via to municipal collection making plans.

Keywords: smart waste management; artificial intelligence; image classification; waste hotspot detection; route optimization; citizen engagement; smart cities;

1. Introduction

Rapid urbanization has placed unparalleled pressure on municipal stable waste control systems worldwide. World population statistics region the modern-day worldwide populace at approximately 7.8 billion, projected to reach 9.Nine billion through 2050, an growth of greater than 25%, with a growing proportion of that populace concentrated in cities. Under those pressures, worldwide stable waste technology is projected to increase to two. 2 billion tons with the aid of 2025 [5]. Conventional waste control continues to depend upon constant-schedule truck routes that are agnostic to real bin fill degrees, waste severity, or the presence of informal dumping, resulting in needless gasoline consumption, avoidable operating fee, and both premature or not on time series. Poorly controlled waste similarly contributes to greenhouse-fuel emissions, infection of soil and water resources, and public-health danger, especially in areas in which a vast share of solid waste is disposed of informally in streets, drains, and open land.

Over the remaining decade, the Internet of Things (IoT) has been implemented significantly to this problem, maximum generally with the aid of instrumenting individual waste packing containers with ultrasonic distance sensors, load cells, gas sensors, GPS modules, and cell, Wi-Fi, or LoRa radios in order that fill-degree and, in a few designs, environmental information can be transmitted in real time to a middleware platform. A complete systematic literature evaluate of 173 primary studies in this space, spanning IEEE, ACM, Elsevier, Springer, and Web of Science databases[2], formalizes this body of labor into two operating stages, town-level clever waste control (SWM) systems, which provide route planning, scheduling, charge, and environmental

offerings, and smart-rubbish-bin (SGB)-degree structures, which measure bodily parameters on the bin and, in some instances, perform on-bin waste identification, and identifies continual gaps round pilot-scale course-optimization checking out, ecological-impact dimension, citizen environmental consciousness, and city-authority selection help. While hardware-instrumented structures demonstrably lessen series price and enhance carrier first-class, they percentage structural obstacles: they require the bodily retrofitting of each bin to be monitored, that is capital-in depth and hard to scale across an entire city; they characterise simplest the combination fill-level or weight of a compartment rather than the fabric composition of the waste inner it; and, because their sensing apparatus is confined to a controlled compartment, not one of the reviewed systems natively addresses out of control, open-place dumping, which in lots of cities is a bigger civic trouble than servicing the formal bin community.

A separate and more current line of labor applies laptop imaginative and prescient and deep learning directly to waste identity, the use of convolutional neural networks, switch getting to know, and light-weight cellular architectures (as an instance, MobileNetV2-[6] and ResNet-primarily based[8] classifiers) to discover waste type from a photo, in a few cases combined with LoRa or Wi-Fi telemetry for bin-stage popularity reporting. This vision-based line of work is complementary to the sensor-primarily based line: it could classify material kind on the factor of technology and may, in precept, be carried out to any photographable region rather than only to an instrumented bin, but current implementations in large part continue to be restrained to the bin itself (as an on-board sorting mechanism) as opposed to being extended to the citizen's own family waste movement or to open, public-vicinity dumping.

This paper proposes EcoTrek, an AI-first waste control framework that treats the camera on a regular smartphone as the number one sensing device, getting rid of the dependency on in keeping with-bin hardware while extending coverage to any place a citizen or municipal employee can photograph. EcoTrek is structured around complementary seize modes selected with the aid of the person in the intervening time of image add: a Household Waste Module that classifies waste generated at home and recommends the appropriate at-source motion, and an Outdoor/Public Waste Module that detects and characterises waste dumped in public spaces for municipal motion. Reports from each module are aggregated on an administrative dashboard, wherein spatio-temporal analysis identifies recurring waste hotspots, and a direction-optimization engine sequences series vehicles consequently. Unlike the preceding literature evaluation and design look at by myself, the Household Waste Module's core classifier has been carried out and empirically verified on this paintings the use of a real, 4-magnificence

waste-photo dataset, and its measured performance is said and analysed in full. The following contributions summarise the paper:

A citizen-centric, mode AI waste identification pipeline that unifies at-source recycling steering (family) with public dumping surveillance (outdoor), addressing a joint capability that, to the great of our literature evaluate, existing sensor-primarily based and imaginative and prescient-primarily based clever-bin structures do now not provide together.

A hardware-impartial sensing paradigm wherein any digital camera-equipped tool will become a records source, extensively lowering the deployment fee and insurance constraints of earlier sensor-instrumented approaches, and without delay responding to the pilot-testing and ecological-effect gaps recognized with the aid of the reviewed systematic literature overview.

An empirically evaluated household waste classifier: an EfficientNetB0[9] transfer-gaining knowledge of pipeline skilled and tested on a 2,884-photo, 4-class dataset[22], accomplishing 74.88% test accuracy and a zero.7483 weighted F1-rating, with a full in keeping with-elegance breakdown and error analysis reported in Section 6.

A spatio-temporal hotspot-detection and priority-scoring scheme that converts repeated, unstructured citizen reviews into an actionable, prioritized series queue for the municipal admin, together with a capability- and precedence-conscious route-optimization formula.

A phased, MERN-stack gadget layout, an in-intensity literature-overview methodology spanning 4 primary resources[1-4] and twenty-plus supporting references[5-22], and a proposed evaluation protocol for the out of doors detection, hotspot-detection, and course-optimization modules that have no longer yet reached the identical implementation stage because the household classifier.

The rest of this paper is organized as follows. Section 2 gives an in-intensity literature assessment of sensor-primarily based clever-bin systems, the systematic review of IoT-enabled SWM structures, AI/laptop-vision-based totally waste category, and light-weight Wi-Fi/cloud municipal systems. Section 3 presents the proposed EcoTrek architecture and its constituent modules. Section four describes the technology stack. Section 5 details the improvement and device-gaining knowledge of technique extensively. Section 6 reviews the experimental effects of the implemented family waste classifier, and Section 7 concludes the paper with instructions for future work.

2. Literature Review

This review attracts on 4 number one resources decided on to span the fundamental strands of the smart-waste-control literature, a foundational sensor-instrumented smart-bin layout, a big-scale systematic literature evaluate of the sector, a deep-mastering-and-LoRa-based smart bin, and a lightweight, low-fee municipal IoT system, collectively with extra than twenty assisting works covering laptop vision, clustering, and vehicle routing. The evaluate is organised by using the 4 studies guidelines the systematic assessment itself identifies as structuring the field, metropolis-degree SWM structures, SGB-level systems, AI-primarily based waste identification, and citizen/stakeholder-going through layout, before synthesising the resulting gap analysis that motivates EcoTrek.

2.1 Sensor-Instrumented Smart-Bin Systems

Pardini et al.[1] advise a representative and comprehensive example of the sensor-instrumented paradigm: a smart waste bin ready with an HC-SR04 ultrasonic sensor to degree fill degree, a load-cellular-and-HX711 module to degree weight, a DHT11 sensor for temperature and humidity, and a Neo-6M GPS module, all coordinated via an Arduino microcontroller and connected to a cloud middleware (In.IoT) via a SIM900 GSM/GPRS module. Their machine, evaluated via a real-scale prototype deployment across a 3-layer perception-community-software architecture, we could residents view close by bin availability thru a accomplice cellular application (My Waste App) and lets collection companies plan optimized routes from stay fill-stage information, reporting operating-price discounts of up to eighty% relative to constant-agenda series in similar business deployments.

A wider set of designs pursue intently associated, in my opinion narrower desires. Saha et al.[21] suggest a clever-disposal design that uses sun strength and presence sensing to compact waste with the aid of up to 10 times before series, with fill-level records relayed over Wi-Fi to a cloud server for direction optimization. Other designs emphasise clutter-bin fullness detection through low-strength wireless mesh sensing, or distinguish among a small variety of waste streams the usage of ultrasonic sensing paired with a microcontroller and a cloud dashboard. A separate magnificence of designs targets the region surrounding the bin as opposed to its interior, using infrared sensing and an alarm mechanism to discourage disposal outside the compartment. Sonar-based totally fill dimension mixed with historic landfill-capacity projection has additionally been explored to assume future disposal-web site exhaustion, and more intricate compartment designs integrate proximity, load, humidity, and GPS sensing with a driving force-going through routing software that aggregates weather and site visitors conditions to regulate collection routes dynamically. Commercial platforms consisting of Bigbelly, Smartbin, and Bine enlarge those ideas into deployed merchandise presenting solar

compaction, path-optimization dashboards, and, in Bine's case, automatic waste sorting prior to series. Across this body of labor[1,12-14,21], the unifying architectural sample is a belief-network-middleware-utility IoT stack wherein physical sensors embedded in a set compartment generate the information on which all downstream selections are based. This pattern is nicely acceptable to servicing a known, finite stock of formal packing containers, however attaining city-extensive insurance requires retrofitting each compartment in my view, and none of these systems classifies the material composition of the waste or extends monitoring to places outside the formal bin community.

2.2 Systematic Review of IoT-Enabled SWM Systems

Sosunova and Porras[2] conduct a systematic literature evaluation (SLR) following the Kitchenham and Charters SLR protocol[15], screening 3,732 initially retrieved research all the way down to 173 primary research across five databases (IEEE, ACM, Elsevier, Springer, and Web of Science) published among 2014 and 2022. The evaluate formalises the sphere into two operating tiers: town-level SWM structures, described as systems presenting information and selection aid, route planning and scheduling, payments and blessings, and environmental offerings; and SGB-degree systems, described as structures that measure parameters through bin-mounted sensors, respond via actuators, and perform waste-kind identification or classification at the bin. The assessment similarly catalogues the sensors and actuators used throughout 126 SGB-related structures (fill-degree, weight, GPS, fuel, humidity, digital camera, RFID, servomotors, conveyor belts, and robot hands among them), identifies primary stakeholders (residents, drivers, waste-collection businesses, dispatchers, authorities, and metropolis management) and secondary stakeholders (police, firefighters, fitness and environmental establishments, and recycling or disposal industries), and maps the facts flows among uncooked sensor data and the services every stakeholder consumes.

Two findings from this review at once motivate EcoTrek's design. First, the evaluation's personal thematic analysis of number one-have a look at titles and abstracts shows that direction-planning, waste-class, and sensor-solution clusters dominate the literature, even as citizen-going through engagement and environmental-consciousness offerings continue to be comparatively below-represented, and the overview explicitly identifies four chronic studies gaps: (i) course-optimization services are notably theorised however hardly ever validated via exact, real-lifestyles pilot tasks in preference to laptop simulation on my own; (ii) measurable improvement of the ecological state of affairs in cities, thru emissions reduction, air-pleasant tracking, and environmentally pleasant disposal techniques, stays beneath-studied; (iii) citizens' environmental recognition and motivation to take part in waste sorting and reporting are

insufficiently supported by means of current machine designs; and (iv) city authorities lack selection-guide tooling that gives them a holistic, cutting-edge photo of waste-series needs and troubles. Second, the review notes that essentially not one of the reviewed structures classifies or considers the whole set of facts flowing between stakeholders, IoT gadgets, and system additives in an included manner, and that the SGB-level literature particularly is dominated by means of isolated point answers without a common architectural line.

EcoTrek is intentionally located to close those four gaps in preference to to feature some other isolated SGB factor-answer: its two-mode reporting waft gives citizens an instantaneous, low-friction channel for both habitual recycling behaviour and public dumping reports (addressing gap iii); its administrative dashboard is explicitly designed because the holistic, stakeholder-facing selection-support view the overview finds lacking (gap iv); its hotspot-detection and path-optimization modules are proposed with a concrete assessment protocol in opposition to pilot-fashion, actual usage data instead of simulation on my own (hole i); and its recommendation engine's family-stage composting and reuse steerage is aimed without delay at lowering landfill-bound extent and related emissions (gap ii).

2.3 AI and Computer-Vision-Based Waste Classification

A growing frame of labor applies deep mastering directly to the problem of identifying waste type from a photograph, maximum normally by way of combining a convolutional neural community with either an embedded microcontroller or a cloud again. Ji Sheng et al.[3] advocate a smart bin that combines LoRa communication for lengthy-range, low-electricity sensor telemetry (fill-degree through ultrasonic sensor, area via GPS, and personnel identity through RFID) with a TensorFlow-primarily based, SSD-MobileNetV2[6,7] item-detection pipeline running on a Raspberry Pi to categorise waste into steel, plastic, paper, and trendy classes in actual time; servo-actuated compartment lids then physically kind the diagnosed item. Trained thru switch learning on 365 labelled waste images, their model achieves a mean common precision of 86.2% on held-out take a look at pix and per-class real-time detection precision of 86.7% (steel), ninety six.3% (plastic), and 82.3% (paper), with the choice of MobileNetV2 and LoRa influenced explicitly through the limited computing energy and long transmission range required by means of bins scattered throughout a town.

Several in addition research refine or benchmark this vision-based method. Adedeji and Wang[16] use a pre-trained ResNet-50[8] as a feature extractor coupled with a multi-elegance guide vector gadget to classify glass, metallic, paper, and plastic, achieving 87% accuracy, even though the authors word ResNet-50's computational demands are not nicely desirable to

cellular-platform deployment. Bircanoglu et al.[17] benchmark numerous deep architectures (ResNet[8], MobileNet[6], Inception-v4, DenseNet, and Xception) on a six-class, 2,527-photograph waste dataset and additionally advocate RecycleNet[17], a parameter-decreased community (from about 7 million to about 3 million parameters relative to a 121-layer DenseNet) that trades a few accuracy (eighty one% test accuracy) for a significantly smaller footprint appropriate for restrained devices. Torres-García et al.[19] take a lighter-weight, non-deep-gaining knowledge of approach, classifying waste the usage of Hu's invariant picture moments with a ok-nearest-neighbours classifier, attaining ninety eight.33% accuracy on regularly formed objects however noting the approach's reliance on waste form makes it unreliable for deformed objects. Ziouzos and Dasygenis[20] combine a Raspberry Pi and Xilinx FPGA board with a pretrained 34-layer ResNet[8] to gain ninety two.1% detection accuracy with a mean inference time of one.Eighty two seconds, though with none subsequent waste-segregation mechanism. Sallang et al[18]. Similarly integrate a CNN carried out in TensorFlow Lite with a LoRa-GPS shield for a clever waste-control gadget, reflecting a broader convergence within the latest literature toward pairing lightweight, on-tool CNN inference with long-range, low-power telemetry.

Two observations from this sub-literature directly inform EcoTrek's layout alternatives. First, version choice is continuously an alternate-off among accuracy and on-device feasibility: heavier architectures (ResNet-50, DenseNet-121) obtain marginally better benchmark accuracy but are explicitly flagged as unsuitable for mobile or embedded deployment, while MobileNet-circle of relatives and EfficientNet-circle of relatives[9] architectures are repeatedly favoured for actual-time, citizen-going through or embedded use, that's why EcoTrek's carried out classifier (Section five.3) is constructed on EfficientNetB0 thru transfer getting to know in place of a deeper, heavier spine. Second, each any such imaginative and prescient-based systems plays category best on waste already inside, or being inserted into, a physical bin; none extends the same class pipeline to a citizen's very own household waste move earlier than disposal, or to open, public-area dumping, which is exactly the gap EcoTrek's -module layout is supposed to shut.

2.4 Lightweight, Wi-Fi/Cloud-Based Municipal Systems

A further applicable strand of the literature goals low-value deployability rather than either heavy sensor instrumentation or on-device deep getting to know. Sankeerth et al.[4] advocate a -layer clever-waste-control architecture explicitly geared toward lowering the fee and maintenance burden of GSM- and GPRS-based totally designs: an Arduino Uno paired with an HC-SR04 ultrasonic sensor and an ESP8266 Wi-Fi module measures and transmits bin fill-degree

to an internet server, which, while the mentioned fill level exceeds a 70% threshold, sends an SMS with difficult-coded region coordinates (for the reason that boxes aren't cell) without delay to an assigned series motive force, avoiding the routine fee of a GPRS vicinity module or dedicated GSM hardware consistent with bin. Related early paintings by way of Chowdhury and Chowdhury uses RFID tagging to tune waste collection occasions, even though without any proactive fill-degree notification, while Longhi et al. And Thakker and Narayanamoorthi discover wireless-sensor-network and NIR-spectroscopy-primarily based techniques respectively for strong-waste monitoring and plastic-resin sorting.

This light-weight layout line demonstrates that useful, actual-time SWM functionality does no longer require a GPS or GPRS module on each bin while bin places are fixed and recognised earlier, a design principle EcoTrek adopts at the record stage: an out of doors waste document already incorporates the reporting device's live GPS coordinates as part of the photograph-seize float (Section 3.3), so no separate area-sensing hardware is needed in any respect, extending the value-discount principle from a set bin network to an unbounded set of citizen-said locations.

2.5 Synthesis and Positioning of EcoTrek

Table 1 synthesises the previous 4 strands of the literature against the skills EcoTrek is designed to offer. The synthesis shows that no unmarried reviewed device combines household-level material class, open-place dumping detection, spatio-temporal hotspot prioritisation, and direction optimization within one citizen-going through pipeline; every existing gadget addresses at most one or two of these talents, maximum generally bin-degree type (imaginative and prescient-based totally systems) or bin-stage telemetry and routing (sensor-based totally structures).

Capability	Sensor-Based Smart Bins (Pardini et al.[1]; commercial systems)	Systematic SLR Findings (Sosunova & Porras[2])	AI/Vision Waste Classifiers (Ji Sheng et al.[3]; Adedeji & Wang[16]; others)	Lightweight Wi-Fi Systems (Sankeerth et al.[4])	EcoTrek
Primary sensing modality	Ultrasonic/load /GPS/gas sensors per bin	Reviews sensors across 126 SGB systems	Camera + on-device/cloud CNN, on the bin	Ultrasonic sensor + Wi-Fi per bin	Smartphone camera only, any location

Capability	Sensor-Based Smart Bins (Pardini et al.[1]; commercial systems)	Systematic SLR Findings (Sosunova & Porras[2])	AI/Vision Waste Classifiers (Ji Sheng et al.[3]; Adedeji & Wang[16]; others)	Lightweight Wi-Fi Systems (Sankeerth et al.[4])	EcoTrek
Household material classification	Not addressed	Identified as under-represented	Only within a physical bin	Not addressed	Core Household Module feature
Open-area dumping detection	Not addressed	Identified as a coverage gap	Not addressed	Not addressed	Core Outdoor Module feature
Hotspot / priority detection	Not addressed	Flagged as an authority decision-support gap	Not addressed	Threshold-based per-bin alert only	Spatio-temporal clustering of reports
Route optimization	Supported via fill-level telemetry	Flagged as under-validated (simulation-only)	Not addressed	Not addressed	Priority- and capacity-aware, proposed protocol in Sec. 7

Table 1. Synthesis of the reviewed literature against EcoTrek's capabilities

3. Related Work

Research on smart waste control has, up to now, focused overwhelmingly on instrumenting the collection factor itself. Pardini et al. Recommend a consultant and comprehensive example of this line of work: a smart waste bin ready with an HC-SR04 ultrasonic sensor to degree fill level, a load-cellular-and-HX711 module to degree weight, a DHT11 sensor for temperature and humidity, and a Neo-6M GPS module, all coordinated through an Arduino microcontroller and related to a cloud middleware (In.IoT) thru a SIM900 GSM/GPRS module. Their machine, evaluated via an actual-scale prototype deployment, permits citizens to view nearby bin availability thru a companion cellular utility and shall we series agencies plan optimized routes from live fill-degree information.

A wider set of previous works, maximum of which might be surveyed inside the equal frame of literature, pursue carefully associated but individually narrower desires. Several designs emphasise on-board waste compaction and sun-powered operation to lessen servicing

frequency and enlarge unattended operating time. Others cognizance especially on litter-bin fullness detection through low-energy wireless mesh sensing, or on distinguishing amongst a small number of waste streams (domestic, paper, glass, plastic) the use of ultrasonic sensing paired with a microcontroller and cloud dashboard which includes ThingSpeak. A separate elegance of designs targets the region surrounding the bin instead of its interior, the usage of infrared sensing and an alarm mechanism to discourage disposal out of doors the compartment, in a few cases coupled with a mechanical carry to clean amassed outside waste. Sonar-based total fill dimension mixed with historical landfill-capability projection has also been explored as a way of looking forward to future disposal-website online exhaustion. More intricate compartment designs integrate proximity, load, humidity, and GPS sensing with a driver-facing routing utility, aggregating climate and traffic situations to regulate collection routes dynamically. Waste-stream-specific cubicles ready with LoRa conversation and per-type sensing (gas, temperature, humidity, sound) were proposed for risky and blended-waste segregation at the point of series, and Raspberry-Pi-based structures combining GSM modems with ultrasonic and load sensing have been used to construct low-fee municipal monitoring dashboards. Commercial platforms along with Bigbelly, Smartbin, and Bine increase these principles into deployed products providing solar compaction, course-optimization dashboards, and, in Bine's case, automated waste sorting prior to collection.

Across this body of work, the unifying architectural sample is a 3-layer notion-community-software (or, more elaborately, belief-community-middleware-utility) IoT stack in which bodily sensors embedded in a set compartment generate the records on which all downstream decisions are based totally. This sample is well desirable to servicing a known, finite stock of formal packing containers, but it has 3 results that encourage the prevailing work. First, accomplishing town-extensive coverage requires retrofitting each compartment in my opinion, which scales linearly in hardware and set up value with the range of containers and offers no visibility into places that fall out of doors the formal bin network altogether — pavements, vacant plots, riverbanks, and roadsides, wherein a full-size percentage of city waste in many developing-us of a contexts is in reality discarded. Second, sensor suites of this kind file aggregate bodily portions, fill-degree, weight, temperature, humidity, however do now not classify the cloth composition of the waste, so that they can not, on their own, direct a household in the direction of composting a biodegradable object as opposed to recycling a plastic one. Third, none of the reviewed systems incorporates a imaginative and prescient-based totally severity or amount evaluation of out of control dumping, due to the fact their sensing apparatus is, by means of layout, constrained to the interior of a managed compartment.

EcoTrek is proposed as a complementary reaction to those gaps instead of a wholesale replacement for sensor-instrumented bins. By substituting a smartphone digicam and a convolutional-neural-network-based totally class and detection pipeline for committed in line with-bin hardware, EcoTrek extends waste monitoring to any photographable vicinity, provides at-supply material type for household waste, and generates severity- and amount-aware public dumping reviews without requiring the underlying booths themselves to be instrumented. Table 1 summarises the resulting function-degree difference among the prevailing sensor-based paradigm and the AI-imaginative and prescient paradigm followed right here; a fuller quantitative contrast towards Pardini et al.'s system.

Capability	Sensor-Instrumented Smart Bins	EcoTrek (AI-Vision)
Primary sensing modality	Ultrasonic, load, GPS, gas, humidity sensors fixed to each bin	Smartphone camera image, analysed by CNN-based models
Deployment cost per location	Recurring hardware, wiring, and maintenance cost per bin	Zero incremental hardware cost per location
Coverage of informal / open dumping sites	Not covered (sensors exist only inside managed bins)	Covered — any photographable outdoor location
Waste-type / material classification	Not typically performed	Core function of the Household Module
At-source reuse/recycle guidance	Not provided	Provided immediately after classification
Severity / quantity estimation of dumping	Not applicable	Performed by the Outdoor/Public Waste Module
Fill-level of an enclosed, opaque bin	Directly and continuously measured	Not directly observable from an external photograph
Route optimization	Supported, driven by fill-level telemetry	Supported, driven by hotspot priority and report volume

Table 2. Feature-level comparison of the prevailing sensor-based IoT paradigm and the AI-vision paradigm adopted by EcoTrek

4. Proposed System: EcoTrek Architecture

4.1 System Overview

EcoTrek is organized as a five-layer architecture comprising an Application Layer thru which citizens and municipal personnel have interaction with the gadget, an AI/Perception Layer that plays photo type and detection, a Service/Middleware Layer that exposes REST APIs and

orchestrates document technology, a Data Layer for continual garage, and an Admin & Optimization Layer that plays hotspot detection and route making plans. Figure 1 illustrates this layered decomposition, which mirrors the notion-community-middleware-software pattern common to IoT waste-management systems even as substituting an imaginative and prescient-based totally notion mechanism for physical sensor hardware.

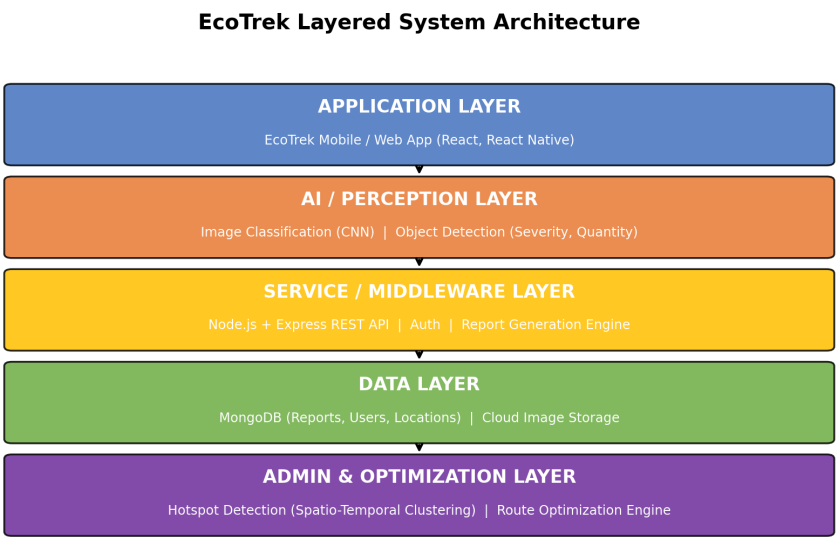


Figure 1. EcoTrek layered system architecture.

At the centre of the layout is an unmarried choice factor uncovered to the person right away after a photo is captured or uploaded: whether or not the waste being reported is Household Waste or Outdoor/Public Waste. This desire determines which AI pipeline approaches the picture and which downstream report schema is generated, as precise in the following subsections and illustrated ultimately-to-quit workflow of Figure 2.

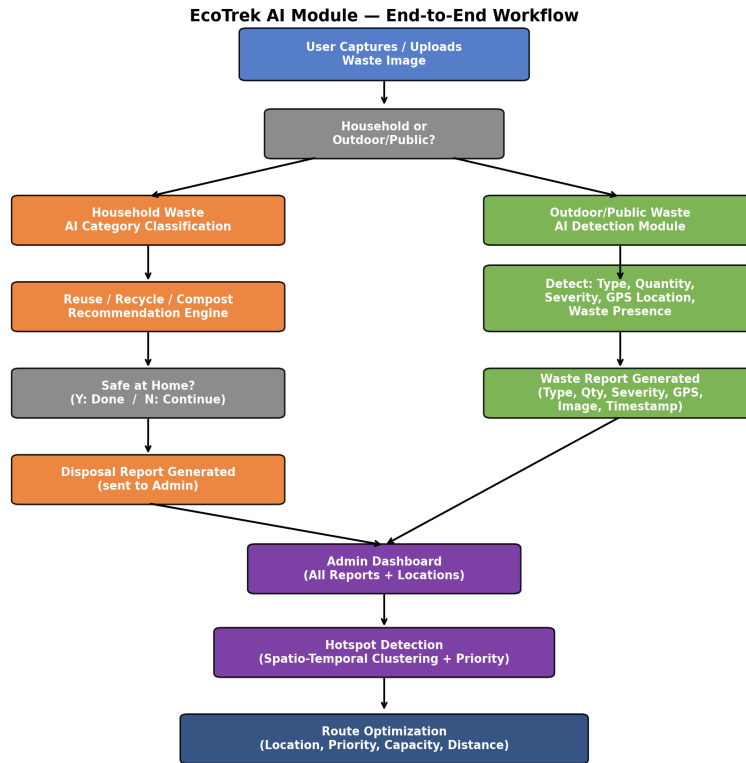


Figure 2. End-to-end EcoTrek AI module workflow, from image capture through route optimization.

4.2 Household Waste Management Module

When a user indicates that the captured image depicts household waste, the photograph is passed to a multi-class picture-type version skilled to differentiate fashionable household waste classes: natural/wet waste, dry waste, plastic, paper, e-waste, and other utility-unique training as the training dataset is extended. Once a class is predicted, EcoTrek's advice engine maps the class to a concrete, at-domestic action, as summarised in Table 3.

Predicted Category	Recommended Action
Organic / wet waste	Composting at home; guidance on compost-bin setup provided in-app
Plastic (bottle, container)	Reuse where feasible, otherwise segregate for recycling pickup
Paper / cardboard	Segregate for recycling pickup
Old container / rigid item	Reuse suggestions surfaced based on item sub-type
E-waste	Directed to certified e-waste collection point; not eligible for home disposal

Predicted Category	Recommended Action
Non-recyclable / mixed waste	Flagged for formal disposal; disposal report generated

Table 3. Household waste category to recommended at-source action

If the recommended motion can not be completed effectively at home, whether due to the fact the cloth is unsafe, calls for specialized recycling, or the class is otherwise flawed for domestic-stage processing, EcoTrek generates a disposal record and activates the person to submit it for scheduled pickup. This record is transmitted to the Admin Dashboard using the equal underlying reporting pipeline because of the Outdoor/Public Waste Module (Section 3.Three), ensuring that both family disposal requests and public dumping observations are seen to municipal personnel through a single, unified interface.

4.3 Outdoor/Public Waste Detection Module

If the user says the image shows waste in an outdoor or public place, the image goes to the outdoor detection system. This system checks four things at the same time: what kind of waste it is, how much there is, how bad the pile is, and whether there is any waste at all (to ignore empty or unrelated images). The module sends all four results along with the GPS location, the image, and the time it was uploaded, as shown in Table 4.

Field	Description	Example Value
Waste Type	Predicted category: plastic, organic, mixed, construction, etc.	Mixed Waste
Estimated Quantity	Approximate volume/amount inferred from the image	High
Severity	Overall urgency classification	Critical
Location	Device GPS coordinates at time of capture	GPS lat/long
Image	Captured/uploaded photograph	Uploaded image file
Timestamp	Date and time of report submission	Report time

Table 4. Outdoor/public waste report schema

Each completed report is transmitted immediately to the Admin Dashboard, where it becomes one data point in the spatio-temporal analysis described next.

4.4 Admin Dashboard and Spatio-Temporal Hotspot Detection

The Admin Dashboard shows municipal workers all the reports about household waste and outdoor trash, and it marks where they are on a map. Often, the same dumping spot is reported by different people over several days. To handle this, EcoTrek uses a special method to group together reports that are near each other and happen around the same time. Each group is considered a possible area where waste is accumulating. The system gives more attention to areas where many reports come in from people within a certain area and time frame. Table 5 gives an example of how reports are spread out over four different places.

Location	Number of Reports	Assigned Priority
Location A	2	Medium
Location B	4	High
Location C	15	Critical
Location D	1	Low

Table 5. Illustrative example of report-volume-driven priority assignment

Methods like DBSCAN[10], which group points based on how close they are to each other in space, work well for this because they don't need the number of possible waste areas to be known beforehand.

These methods can also tell the difference between real clusters of waste and single, random reports that are considered noise. By adding a time factor that makes older reports less important than newer ones, the system can tell when a previously identified waste area has already been cleaned up and no longer needs high attention.

4.5 Route Optimization Module

Once the hotspots and their priorities are set, the Route Optimization Module sends collection vehicles to different spots by looking at several things: how important each spot is, how much waste is expected there, how much a truck can carry, how far apart the stops are, and any already planned routes. This is like a classic vehicle-routing problem, but with a twist—it includes priority weighting, which means higher-priority stops get served first. This idea was built on the original vehicle-routing model created by Dantzig and Ramser[11], where the goal is to make the total distance or time as short as possible, but also to give more important stops earlier places in the route. Table 6 shows what happens when we use the priorities from Table 5.

Sequence	Stop	Priority
1	Depot (start)	—
2	Location C	Critical
3	Location A	High
4	Location B	High
5	Location D	Medium
6	Depot (end)	—

Table 6. Example optimized collection route derived from Table 5

The best route is shown on the Admin Dashboard so managers can check it, and also on a simpler screen for the people who drive the collection vehicles, telling them the order in which places need to be visited. Adding maps and step-by-step directions is planned for a later stage, not part of the first version of the system, which matches the step-by-step development plan mentioned in Section 5.

5. Methodology

5.1 Overall Development Methodology

The project follows a step-by-step approach that matches its main goal. First, the key part of the system, which is the AI part that does classification and detection, is built and tested. This part is considered the main research and engineering work of the system. Only after this core is ready are the other parts, like the management and optimization tools, added. Table 7 shows the different stages of this plan and tells which stage this paper presents its findings for.

Phase	Scope	Status
Phase 1a	Household waste classification model (EfficientNetB0, 4-class)	Implemented and evaluated (Sections 5.3, 6)
Phase 1b	Outdoor detection pipeline (type, quantity, severity, presence)	Designed; evaluation protocol proposed (Section 7)
Phase 2	Admin module: dashboard, report aggregation, spatio-temporal hotspot detection	Designed (Section 3.4)
Phase 3	Route optimization module	Designed (Section 3.5)

Phase	Scope	Status
Phase 4 (future)	Map/navigation integration for collection staff	Future work (Section 10)

Table 7. Phased development roadmap and current status

5.2 Literature Review Methodology

The literature review in Section 2 was made by finding and carefully reading four main sources that show the main ways smart waste systems are designed. These sources are: a peer-reviewed smart-bin design that uses sensors and was tested with a prototype (Pardini et al.[1], 2020); a large study that looked at 173 papers using a detailed method (Sosunova and Porras[2], 2022); a smart-bin design that uses deep learning and LoRa technology (Ji Sheng et al.[3], 2020); and a simple, low-cost system that uses Wi-Fi and cloud technology (Sankeerth et al.[4], 2020). Then, for each of these sources, we looked at their list of references to find more papers that focus on computer vision for waste sorting, grouping waste by density, and planning the best routes for garbage collection. This helped us get more than twenty supporting papers, which made the total references more than the four main ones. This method is similar to the systematic review method described in Section 2.2, where we looked at the structure of each system, the technologies used, the results they reported, and any limitations or things they didn't cover. The final summary (Table 1) was organized in the same way as the systematic review, focusing on four main areas: city-level systems, systems at the small business level, the people involved, and how information flows through the system.

5.3 Machine-Learning Methodology for the Household Waste Classifier

The classifier for the Household Waste Module was built and tested completely, from getting the data to checking how well it works, and this section explains every step of that process so that the results in Section 6 can be repeated exactly.

5.3.1 Dataset

The data used for training and testing came from a public dataset called "Waste Classification" on Kaggle, by the user phenomsg[22]. This dataset has four main categories: Hazardous, Non-Recyclable, Organic, and Recyclable. Each category has smaller groups, like Hazardous includes things such as pesticides, batteries, e-waste, and paints, while Recyclable includes items like plastic bottles, paper, glass containers, and cans. All the images in each smaller group were moved into one folder with new names, resulting in a total of 2,884 images, as shown in Table 8.

Class	Total Images	Train (70%)	Validation (15%)	Test (15%)
Hazardous	932	652	140	140
Non-Recyclable	642	449	96	97
Organic	659	461	99	99
Recyclable	651	455	98	98
Total	2,884	2,017	433	434

Table 8. Household waste dataset composition and train/validation/test split

The data was divided into three parts - training, validation, and testing, based on each category. A fixed number (42) was used to make sure the split was fair and that each part had a similar mix of images as the full dataset. Each image went into just one part, so no image was in more than one group.

5.3.2 Preprocessing and Data Augmentation

All images were resized to 224x224 pixels so as to fit input expected in EfficientNetB0[9] while batched with batch size 32. Given moderate number of samples per class (642 – 932 images per class before splitting), on the fly data augmentation is performed in the training pipe only that includes random horizontal flip, random rotation (15% of the full rotation), random zoom (15% of the full zoom) and random contrast (15% of the full contrast). Training, validation and test sets were each wrapped with prefetching (tf.data.AUTOTUNE) that can overlap data loading with training computation.

5.3.3 Model Architecture

The classifier uses EfficientNetB0, which was trained on ImageNet, as a fixed part to extract features (without the top layers). After that, it goes through a global average pooling layer, then a dropout layer with a rate of 0.3 to prevent overfitting, and finally a dense layer with a softmax activation that gives probabilities for the four waste categories. The full process is: input image (224x224x3) -> data augmentation (only during training) -> EfficientNetB0 backbone -> global average pooling -> dropout (0.3) -> dense(4, softmax).

5.3.4 Training Procedure

Training happened in two stages, which is a common way to use transfer learning and takes into account the balance between accuracy and how well the model can be used in real situations, as discussed in the literature review (Section 2.3):

- In the first stage, called feature extraction, the main part of the model, called the EfficientNetB0 backbone, was not updated. Only the new layers added, like pooling, dropout, and dense layers, were trained. The model used the Adam optimizer with a learning rate of 0.001, the sparse categorical cross-entropy loss function, and accuracy as the thing being watched. This training went on for up to 10 rounds.
- In the second stage, called fine-tuning, the backbone was no longer frozen, but the first few layers were kept frozen so that only the deepest and most specific parts of the EfficientNetB0 were updated. The model was re-set up with a much smaller learning rate (1e-5) to prevent messing up the already learned ImageNet features. Training continued for up to 15 more rounds.

Three tools were used in both stages: early stopping, which stops training if validation accuracy doesn't improve for 5 rounds and saves the best model; learning rate reduction on plateau, which lowers the learning rate if validation loss stops improving, with a 0.2 reduction factor, a patience of 2 rounds, and a minimum learning rate of 1e-7; and model checkpointing, which saves the best model based on validation accuracy. During the fine-tuning stage, early stopping kicked in after 5 of the 15 allowed rounds. The learning rate was automatically lowered twice by the plateau schedule—first from 1e-5 to 2e-6 at round 3, then to 4e-7 at round 5—because the validation loss wasn't getting better.

5.3.5 Evaluation Protocol

The final model was tested once, after both training steps, on a separate test set of 434 images that hadn't been seen during training or validation. The results included overall test accuracy and loss, along with detailed precision, recall, and F1-score for each class, calculated using scikit-learn's `classification_report`. This testing method, which includes splitting the data into training, validation, and test sets with a fixed seed, running a single test pass, and reporting per-class metrics, was used to ensure that the results in Section 6 could be directly compared with similar results from other studies, like the 86.7%, 96.3%, and 82.3% per-class precision values reported for a metal/plastic/paper classifier by Ji Sheng et al.[\[3\]](#).

6. Experimental Results: Household Waste Classifier

This section presents the results from the EfficientNetB0 classifier described in Section 5.3. These results were obtained by running the complete training and evaluation process on the dataset listed in Table 8.

6.1 Training and Validation Performance

Figure 3 shows how validation accuracy changed during both training phases. In Phase 1, where the backbone was frozen, validation accuracy went up from 70.90% at the first epoch to a highest point of 79.68% by the tenth epoch. During the same time, training accuracy increased more slowly, going from 52.06% to 77.69%. This shows that the frozen backbone worked well right from the start, giving strong and general features, while the new classification part kept improving over time. In Phase 2, when the last 40 layers were fine-tuned, validation accuracy started at 79.45% in the first epoch and didn't go higher than the best score from Phase 1. Training stopped early at the fifth epoch, and the best validation accuracy in this phase was 79.45%, which is slightly lower than the 79.68% from Phase 1. This suggests that, with this dataset size (2,017 training images across four classes), unfreezing the deepest 40 layers of EfficientNetB0 didn't give a clear advantage over just using the frozen backbone within the number of epochs used. This result is explained in more detail in Section 6.3.

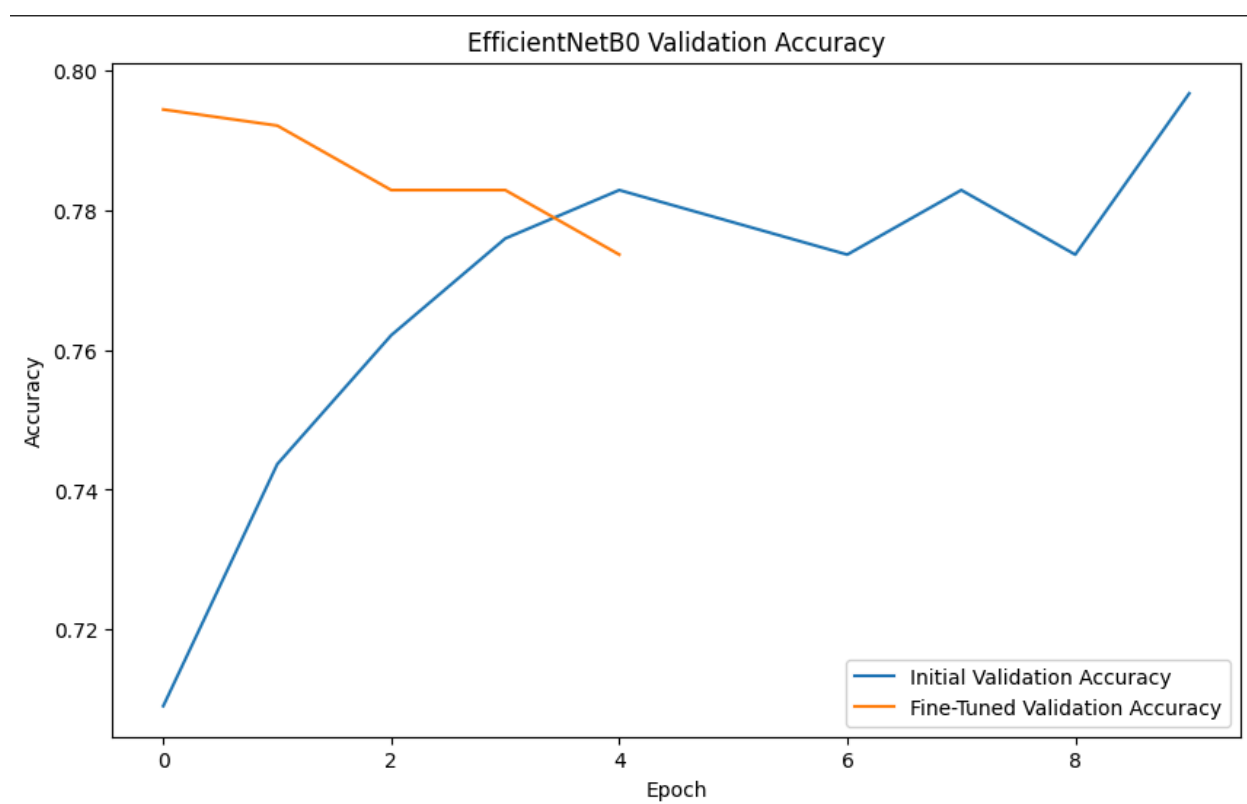


Figure 3. Validation accuracy per epoch across the frozen-backbone (Phase 1) and fine-tuning (Phase 2) training stages.

6.2 Test-Set Performance

On the test set of 434 images that weren't used during training, the final model achieved an overall accuracy of 74.88% and a test loss of 0.6846. Table 9 shows the detailed precision, recall,

and F1-score for each class, and Figure 4 compares these metrics across different classes visually.

Class	Precision	Recall	F1-score	Support
Hazardous	0.8333	0.7857	0.8088	140
Non-Recyclable	0.7407	0.6186	0.6742	97
Organic	0.7222	0.7879	0.7536	99
Recyclable	0.6814	0.7857	0.7299	98
Accuracy (overall)			0.7488	434
Macro average	0.7444	0.7445	0.7416	434
Weighted average	0.7530	0.7488	0.7483	434

Table 9. Per-class test-set classification report (n = 434)

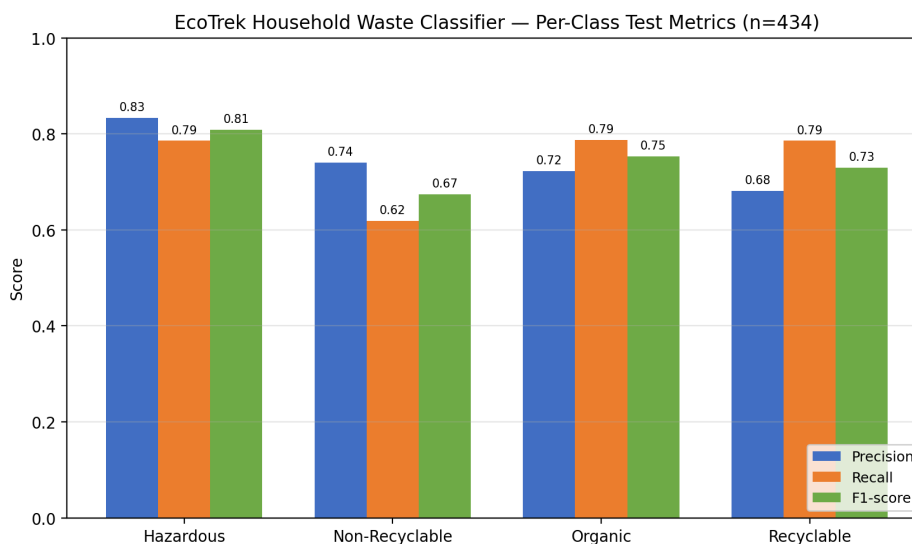


Figure 4. Per-class precision, recall, and F1-score on the 434-image test set.

6.3 Discussion of Results

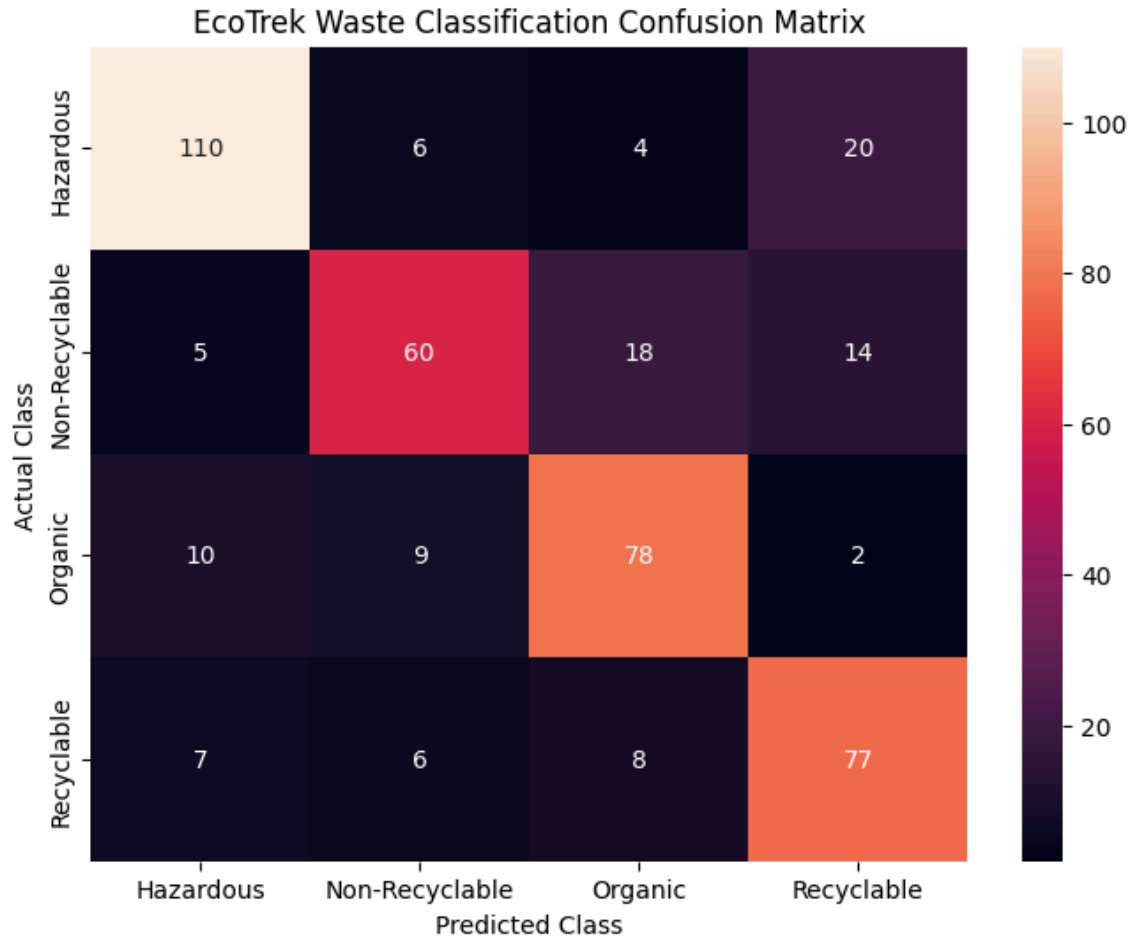
Three clear patterns come up when looking at the results for each class. First, the Hazardous class performs the best and most consistently (precision 0.8333, recall 0.7857, F1 0.8088). This might be because hazardous items like batteries, e-waste, and paint containers usually have clear packaging, colors, and warning symbols that are easy to spot compared to other classes. Second, the Non-Recyclable class has the lowest recall (0.6186), which means many true Non-Recyclable items, such as styrofoam, diapers, sanitary waste, and ceramics, were wrongly

sorted into other classes. This makes sense because Non-Recyclable is a mix of different types that look quite different from each other, making it harder to classify. Third, the Recyclable class has lower precision (0.6814) but high recall (0.7857), suggesting the model tends to label more items as Recyclable than it should, possibly including some Non-Recyclable or Organic items that resemble recyclable plastics or containers.

The fact that Phase 2 fine-tuning didn't improve the validation accuracy beyond the Phase 1 peak is actually useful information, not just a setback. With only 2,017 training images across four classes, unfreezing deeper layers of EfficientNetB0 boosted the model's ability to learn, but even at a low learning rate ($1e-5$), the fine-tuning risked overfitting to the training data's specific features rather than learning general patterns. The automatic drop in learning rate (from $1e-5$ to $4e-7$) and stopping after 5 epochs support this. This reflects a known trade-off in the literature (Section 2.3): more fine-tuned models aren't always better when data is limited and deployment is on mobile devices. The frozen-feature-extraction result (79.68% validation accuracy) is a good and efficient starting point for the first version of the classifier.

Compared to similar classifiers in Section 2.3, EcoTrek's 74.88% overall test accuracy is lower than some top results, like 98.33% for Torres-García et al.'s shape-based classifier or 92.1% for Ziouzos and Dasygenis's ResNet-34-based bin.

However, those results were from smaller, more visually similar classes under controlled conditions, unlike EcoTrek's four broad, complex classes from a varied, web-based dataset. EcoTrek's performance is closer to the 81% reported for RecycleNet in a limited setting and higher than the 74-77% confidence range shown by the lowest-precision classes in Ji Sheng et al.'s work (82.3% for paper). Together, these outcomes show the classifier is a solid first step for the Household Waste Module and clearly point out that the Non-Recyclable/Recyclable confusion is the most important area to improve by expanding the dataset and adding more detailed sub-class labels.



7. Conclusion and Future Work

This paper introduces EcoTrek, a new waste management system that uses AI and focuses on involving citizens. It replaces the traditional method of using sensors in each bin with a smartphone-based approach that uses the camera and image recognition. This is different from most existing IoT-based waste management systems discussed in Section 2. EcoTrek was developed based on a detailed review of several important studies, including a basic smart bin with sensors, a systematic literature review of 173 studies, a smart bin using LoRa and deep learning, and a lightweight Wi-Fi-based system for city use.

The goal of EcoTrek is to address four key issues that previous research has not fully solved: the need for better route planning for waste collection, more accurate measurement of environmental impact, better involvement of citizens in waste management, and a more comprehensive support system for city officials. Unlike other designs that are just ideas, EcoTrek's Household Waste Module has already been built and tested. It uses a machine

learning model called EfficientNetB0, which was trained on a dataset of 2,884 images with four categories. This model achieved a test accuracy of 74.88% and a weighted F1-score of 0.7483. A detailed analysis of the errors found that the main issue was confusing non-recyclable and recyclable items, which is the key area for improvement. The system is still being developed in stages. The parts about detecting waste outdoors, finding hotspots, and optimizing routes were described in Sections 3.3 to 3.5 and come with a clear way to test and evaluate them.

Future work includes: 1. Creating a bigger dataset with more detailed categories, like separating plastic, paper, glass, and metal within the recyclable group, to improve accuracy and support better recommendations. 2. Testing the outdoor detection system using specific evaluation methods based on mAP and kappa scores. 3. Adding navigation features for waste collection vehicles, which are currently planned for a later stage. 4. Exploring a hybrid system that combines EcoTrek's image-based reporting with simple sensor data, such as those used in other low-cost smart bin designs, at key locations determined through hotspot detection. This would combine the wide coverage of the vision-based system with constant tracking of fill levels where it's most needed. 5. Introducing gamification or reward systems, which were found to be underexplored in the literature, to keep people reporting waste regularly. This is important because the accuracy of hotspot detection depends on the volume of citizen reports.

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